Addis Ababa City Administration Educational

Bureau Special Tutorial Program

Condensed Biology Curriculum for Grade 11

StudentsDesigned From Grades 9-11

September 2026 G.C
Addis Ababa/Finfinne

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Preface

Dear teachers,

Addis Ababa City administration Education Bureau has prepared this condensed Biology curriculum highlighting on the core lesson notes and selected some key questions based on the syllabus and past exam papers. In addition, this special condensed curriculum contains contents and activities for delivered in 4 days lesson.

As a professional, feel free to modify, add extra explanations or include examples that you think will help students especially, if you know where the students usually struggle. Moreover, you can bring in your own teaching experiences; employ different techniques to keep the students engaged for the 4-hour tutorial classes.

Generally, the program emphasizes ensuring students' engagement in ample practices and completing the sessions with greater confidence that would enhance their readiness for the university entrance examination.

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Lesson One

Time Allowed: 1hr

Topic: Plants

Sub-topic: Characteristics of Plants

Learning competencies: **By the end of this section, students will be able to**:

- Explain the characteristics of plants
- Differentiate between flowering and non-flowering plants
- Draw and label the internal and external structures of plants
- Draw and label the reproductive cycles of flowering and non-flowering plants using typical examples.

The following learner-centered methods are Suggested Teaching Methods:-

- Observation method
- Activity-based Learning- examining roots, stems, leaves
- Demonstration method
- Questing-and- Answer Method
- Group discussion

Questions:

- What are the characteristics of plants?
- How can you prove that a plant is alive?

The word plants include a range of living organisms which share many characteristics in common.

Some of their characteristics are common with other living things and they have some unique characteristics.

Generally plants have the following characteristics:

Plants are living things: like other living things plants can grow, reproduce, and respond to environmental changes.

Plants are multicellular: they are made up of many eukaryotic cells.

Plants are autotrophic (**Self-feeding**): Plants can make their own food by the process of photosynthesis. They are also known as **producers.**

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Plants are Sessile: They cannot move by themselves. They remain fixed at one place, firmly attached to the soil by their root.

However, plants still show movements or responses without moving from one place to another.

Example: Growing towards light (Phototropism)

Root growing downward in response to gravity (geotropism) …etc.

Plants Practice asexual and sexual reproduction patterns: In lower plants (such as mosses and liverworts) asexual reproduction through spores is the dominant form. In higher and seed-bearing plants such as gymnosperms and angiosperms involves the union of gametes or sex cells -sexual reproduction is dominant.

Plants respond to environmental stimuli: They are sensitive to light, water, touch...

Flowering and Non-flowering Plants

Plants are widely associated with those that have flowers, roots and leaves having transporting vessels (vascular bundles). However, there is a huge number of lower plants such as mosses and liverworts which are non-vascular, i.e, do not have transporting system or conducting vessels, essential to transport water, nutrients and food needed for the plant.

The non-vascular plants are generally small with limited height and restricted to moist and shaded areas. These lower plants are seedless and do not have flowers and fruits.

There are still large numbers of plants known as gymnosperms with well-developed root stem and leaves but have no flowers.

Their (gymnosperms') reproductive structure is cone. The seeds produced in their cone are without cover (naked seeds).

Flowering plants commonly known as Angiosperms are also vascular with well-developed root, stem, and leaves. But unlike gymnosperms, they have flowers and produce seeds within a fruit.

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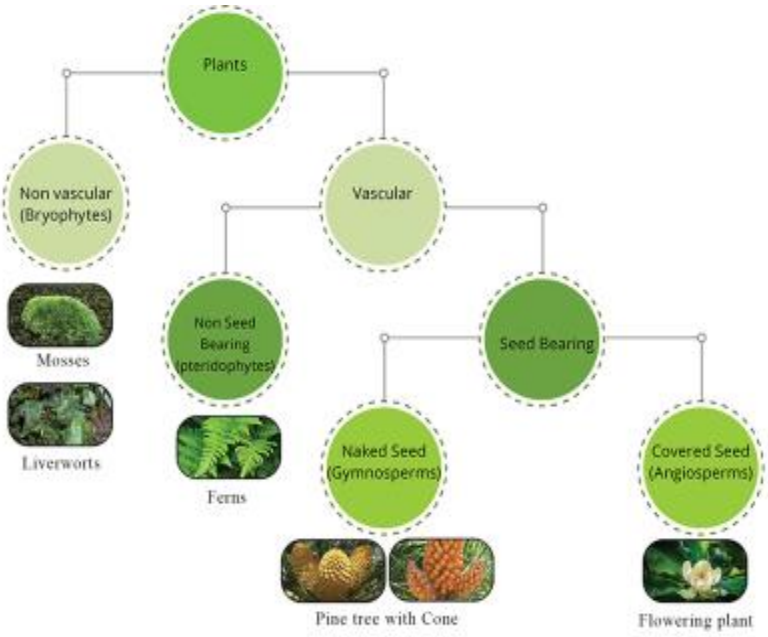


Figure 2.1 Major groups of plants

Exercise

1. A student places a potted plant near a window. After several days, the stem bends toward the sunlight. Which characteristic of plants does this demonstrate?

A. Reproduction B. Response to stimuli C. Excretion D. Locomotion

2. A farmer observes that a bean plant remains attached to the soil throughout its life. However, its shoots and leaves continue to grow and respond to the environment. This shows that plants are generally.

A. Autotrophic B. Multicellular C. Motile D. Sessile

3. A gardener cuts a stem from a rose plant and uses it to grow a new rose plant. The new plant develops without the formation of seeds. This demonstrates:

A. Asexual reproduction B. Sexual reproduction C. Respiration D. Photosynthesis

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Lesson Two

Time Allowed: 1hr

Sub-topic: Structure and Function of Plant Parts

Objectives: At the end of this section, the student will be able to:

- Draw and label the external and internal structures of angiosperms.
- Relate the structure of plants with the function they perform

Guiding Questions

1. What are the major structures of plant?

2. Which part of plant found below the soil?

3. List the two common systems in angiosperm plants?

The external structure of a typical angiosperm has two major systems

1.The Shoot System:

The shoot system is the part of a plant that usually grows above the ground.

It develops from the Plumule of the embryo during germination. The main parts of shoot system are: stem, leaves, flowers, buds, fruits. Flowers and fruits may be missing depending on the reproductive stage of the plant.

2. The Root System: The part of a plant that usually grows below the ground.

Develops from the radicle of the embryo during germination

It consists of main roots, lateral roots, root hairs, and root cap.

Roots don't bear either leaves or buds.

Types of Root

Tap Root

- ✓ Has one large main root
- ✓ Smaller lateral roots grow from it
- ✓ By penetrating deep into the soil, tap roots provide stability (anchorage) and absorb water located deep in the ground.
- ✓ Tap root system is a feature of dicots.

Examples: carrot, pea, mango…

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Fibrous Root

- ✓ Usually formed by thin, moderately branching roots growing from stems. They are more or less similar size and length.
- ✓ No single main root is dominant
- ✓ Very efficient for absorbing water and minerals close to soil surface
- ✓ It creates a thick network of roots that are good at holding soil together and protect soil from erosion.
- ✓ Fibrous roots are features of monocot plants.

Example: Grass, maize. Wheat…

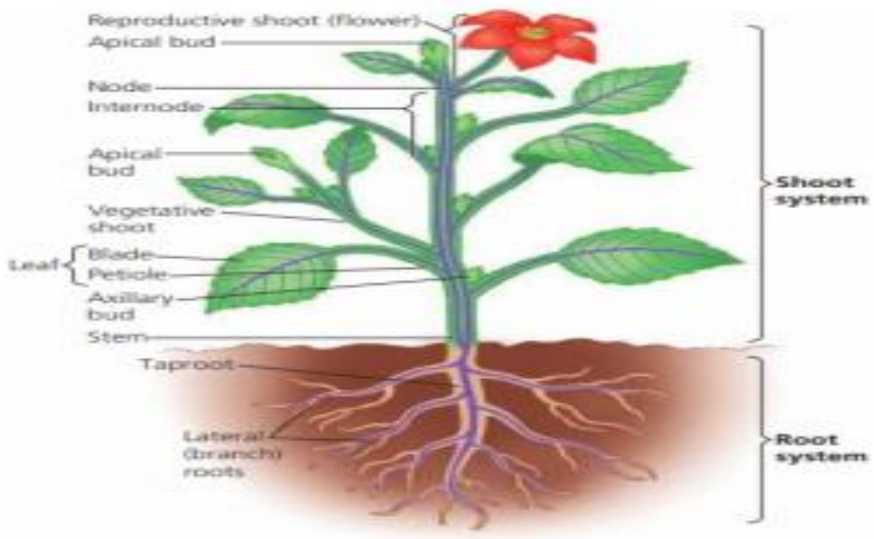


Figure 2.2 External structure of a typical angiosperm (flowering plant)

External and Internal Structures of Leaf

External Structures

A typical leaf consists of the following external structures. These structures are adapted to help the leaf perform its function especially photosynthesis, transpiration….

Petiole

- ✓ The stalk that connects the leaf blade (lamina) to the stem
- ✓ Supports the leaf
- ✓ Contains tissues that transport water, minerals, food between the leaf and stem
- ✓ Position the leaf to receive maximum sunlight

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Lamina (Leaf blade)

- ✓ Broadest, flat and green part
- ✓ Usually the largest part.
- ✓ Provide a large surface area to capture more sunlight

Mid-rib

- ✓ The large central vein running straight down the middle of the leaf blade from the base to the tip
- ✓ Harder and contains the vein (transporting vessels) of the leaf as well as supporting tissues with hard cell wall
- ✓ Provides rigid support and channels water and nutrients

Leaf margin: The outer edge or boundary of the leaf blade

Leaf apex: The terminal tip at the far end of the leaf blade

Question: A student compares two leaves, Leaf A has a broad, flat lamina, while Leaf B has a very narrow lamina. Both receive the same amount of sunlight.

Which advantage does leaf A generally have?

A. Greater surface area for capturing light C. Greater ability to transport water through roots

B. Greater ability to absorb minerals from soil D. Greater production of root hairs

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Lesson Three

Duration: 1hr

Sub-topic: Internal Structures of Leaf

Objective: **Relate the internal structures with their function they perform**

Suggested Teaching Methods

Demonstration: show a real leaf demonstration and demonstrates how a leaf cross-section is arranged

Diagram-based Teaching: Use a clearly labeled transverse section diagram and explain each layer.

Cooperative learning

A leaf is a major photosynthetic organ of a plant. Its internal structure is highly adapted for:

- Photosynthesis
- Gas exchange
- Transport of water and minerals
- Transport of manufactures food
- Control of water loss

Components of the two internal layers of a leaf, namely; Outer layer and middle (inner) layer

A. **Outer layer**: This is also known as the epidermis, a single layer of tightly packed cells that covers the upper and lower surface of the leaf.

The upper epidermis is usually covered by cuticle.

Cuticle: it transmits sunlight for photosynthesis but restricts water loss by evaporation.

Question 1 A student removes the waxy cuticle from the upper epidermis of a leaf. After several hours, the leaf begins to wilt. Why does removing the cuticle causes the leaf to wilt?

A. It prevents photosynthesis B. It prevents mineral absorption

C. It stops food transport D. It increases water loss from the leaf

The lower epidermis usually contains bean-shaped guard cells that leave open spaces known as **stomata.**

Stomata: "little mouths" or "little noses" which regulate O_2 release and CO_2 intake and water loss

In most leaves stomata are more abundant in the lower epidermis, reducing water due to direct sunlight.

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Question 2: A biology teacher places two identical leaves in sunlight. In one leaf, the lower epidermis is coated with petroleum jelly. In the outer, the upper epidermis is coated with petroleum jelly.

Predict which leaf is likely to show a greater reduction in gas exchange and explain your reasoning.

Middle layer

This is known as the mesophyll (middle leaf). It lies between the upper and lower epidermis.

It includes tissues that directly or indirectly involved in photosynthesis.

There are two regions in the mesophyll layer.

The Palisade layer

- Composed of regularly arranged and closely packed columnar (vertically elongated) cells.
- The cells contain the largest number of chloroplast per cell.
- The layer is immediately beneath the upper epidermis
- It is the best position to capture most of the sunlight and enables it to carry out most of the photosynthesis
- The slight but precise separation of the columnar cells maximizes the diffusion of CO_2 and capillary movement of H_2O.

The Spongy layer

- Lies below the palisade cells
- Irregularly shaped
- Fewer chloroplasts
- Very loosely arranged with numerous airspaces. These air spaces, which are very close to the stomata allow the diffusion of O_2 water vapour and CO_2

Guard Cells

- Specialized epidermal cells that surrounds each stomata
- Control whether the stomata pore is opened or closed.
- Contains few chloroplasts

Question 3: A student observes a leaf under a microscope and finds that the cells just beneath the upper epidermis are elongated and contain numerous chloroplasts.

Which function is most strongly associated with these cells? A. Transpiration B. Absorbing light for photosynthesis C. Controlling water loss through stomata D. Transporting minerals to the roots

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Question 4: A disease destroys many cells of the spongy mesophyll but leaves the palisade layer intact.

Which process is most likely to be directly reduced? A. Opening of stomata B. Internal diffusion of gases C. Transport of water D. Prevention of water loss

Question 5: When you look at the transverse section of a leaf under a microscope, which of the following is the correct sequence from the upper surface to the lower one?

A. Palisade cell → spongy cells → upper epidermis → lower epidermis

B. Palisade cell → upper epidermis → lower epidermis spongy cells

C. Upper epidermis → spongy cells → palisade cell → lower epidermis

D. Upper epidermis → palisade cell → spongy cells → lower epidermis

Lesson Four

Duration: 1hr

Sub-topic: Internal structures of a stem and root

Objective: Explain the structures with their function they perform

Suggested Teaching Method

- **Demonstration:**
- **Diagram-based Teaching**:
- **Cooperative learning**

The Internal Structures of Stem

Question 6: What are the internal structures of stem?

The detailed internal structures of stem include:-

The Epidermis

- The outermost layer of the stem
- The outer walls are greatly thickened with cuticles
- Minimizes the rate of transpiration
- Cells are compactly arranged
- Protect the underlying tissues from mechanical injury and prevent the entry of harmful organisms.

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Hypodermis

- Lies below the epidermis
- Mainly composed of collenchyma cells that are specially thickened at the corners due to the deposition of thick cellulose. This enables the layer to give mechanical strength to the stem

Cortex

- Consists of few layers of thin-walled, large, round or oval cells
- Having intracellular space and serving for storage of food

Endodermis

- The innermost layer of the cortex
- Separates the cortex from the vascular bundles
- The cells are compactly arranged and usually contain starch grains
- Endodermis serves as a food reserve and may be termed as a starch sheath.

Vascular bundles

- Longitudinal strands of conducting tissue or transporting vessels
- Consisting essentially of xylem and phloem arranged in a ring around the central pith.
- Each bundle has a patch of xylem towards the pith and a patch of phloem towards the endodermis and a strip of actively dividing young cells (cambium) in between them.

Xylem: transports water and dissolved minerals to the photosynthetic tissues,

Phloem: transports synthesized food to different tissues, either for utilization or storage

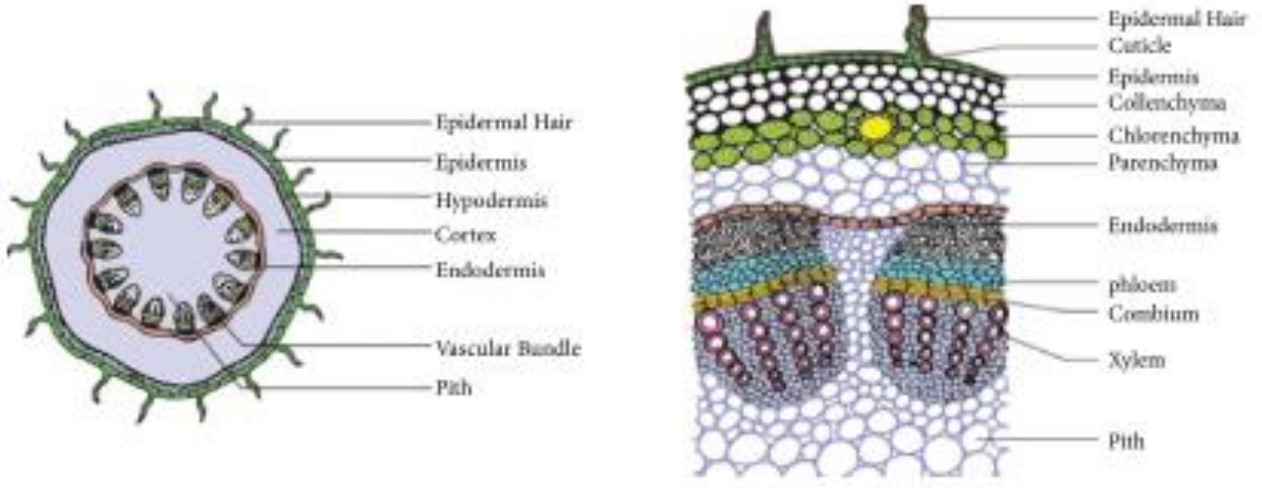


Figure 2.5 Internal structure of a typical dicot stem (A = Ground plan; B = Transverse section)

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Question 7: A researcher damages most of the cortex tissue of a young stem.

Which function is most likely to be affected? A. Storage and movement of substances through the stem B. Production of pollen grains C. Absorption of water from soil D. Opening and closing of stomata

Question 8: A ring of tissue containing the phloem is removed from around a stem while the xylem remains intact. What is most likely to happen below the damaged region? A. Photosynthesis stops immediately B. Roots receive more sugars C. Water transport completely stops immediately D. Sugars accumulate above the damaged region

Question 9: This item is based on the following list of information about route of transport of water in plants

1. Entry of water from stem xylem into leaf

II. Entry of water into the root xylem

III. Flow of water through xylem tissues in stems

IV. Entry of water from soil into root hairs

V. Evaporation of water from leaf surfaces into the environment

Which one of the following alternatives show the correct route of water in plants?

a. V → I → III → II → IV c. V → III → II → I → IV

b. IV → III → II → I → V d. IV → II → III → I → V

Question 10: This item is based on the following table that shows the structure of plants and their corresponding functions.

No	Plant structures	Functions
1	Epidermis	Reduces water loss through transpiration
2	Endodermis	Produces food by photosynthesis
3	Vascular bundles	Conduct nutrients to different part of the plant
4	Cortex	Prevent tissues from infection by pathogens

Which alternative contains the numbers with the correct plants structure and its function?

a.1 and 4 b. 2 and 3 c. 1 and 3 d. 2 and 4

The Internal Structures of Root

The internal structure of a root can be understood as layers arranged from the outside toward the center.

A typical root consists of: **Peliferous layer**, **Cortex**, **Endodermis**, **Pericycle**, **Vasculartissue**, **pith**

Peliferous layer

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- The outermost layer made up of single-layer cells.
- The cuticle is absent
- Consists the single-celled root hairs

Cortex

- Multi-layered zone made of thin-walled oval or rounded loosely arranged cells with intracellular spaces.
- Stores food and water

Endodermis

- The innermost layer of the cortex
- Made up of barrel-shaped closely packed cells
- The layer helps the movement of water and dissolved nutrients from the cortex into the xylem

Pericycle

- A single layer inner to endodermis
- It is the site of origin of lateral roots

Vascular bundles

- Consists of xylem and phloem with meristmetic (cambium) or actively dividing cells between them.

Pith

- Present in young roots while absent in cold roots

Figure 2.6 Internal Structure of root

Question 11: What is/are difference between arrangements of internal structures in monocot> and dicots?

Question 12: A chemical damages the endodermis of a root. Why could this seriously affect water and mineral uptake into the vacuolar tissue? A. The endodermis helps regulate movement into the vascular cylinder

B. The endodermis produces root hairs C. The endodermis carries sugars to the leaves D. The endodermis is the main photosynthetic tissue

Question 13: A researcher observes that new lateral roots develop from a layer immediately inside the endodermis.

Which tissue is most likely involved? A. Epidermis B. Pericycle C. Cortex D. Pith

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